Assignment No 2

Title: Develop a basic facial detection system using TensorFlow.js to detect faces in an image.

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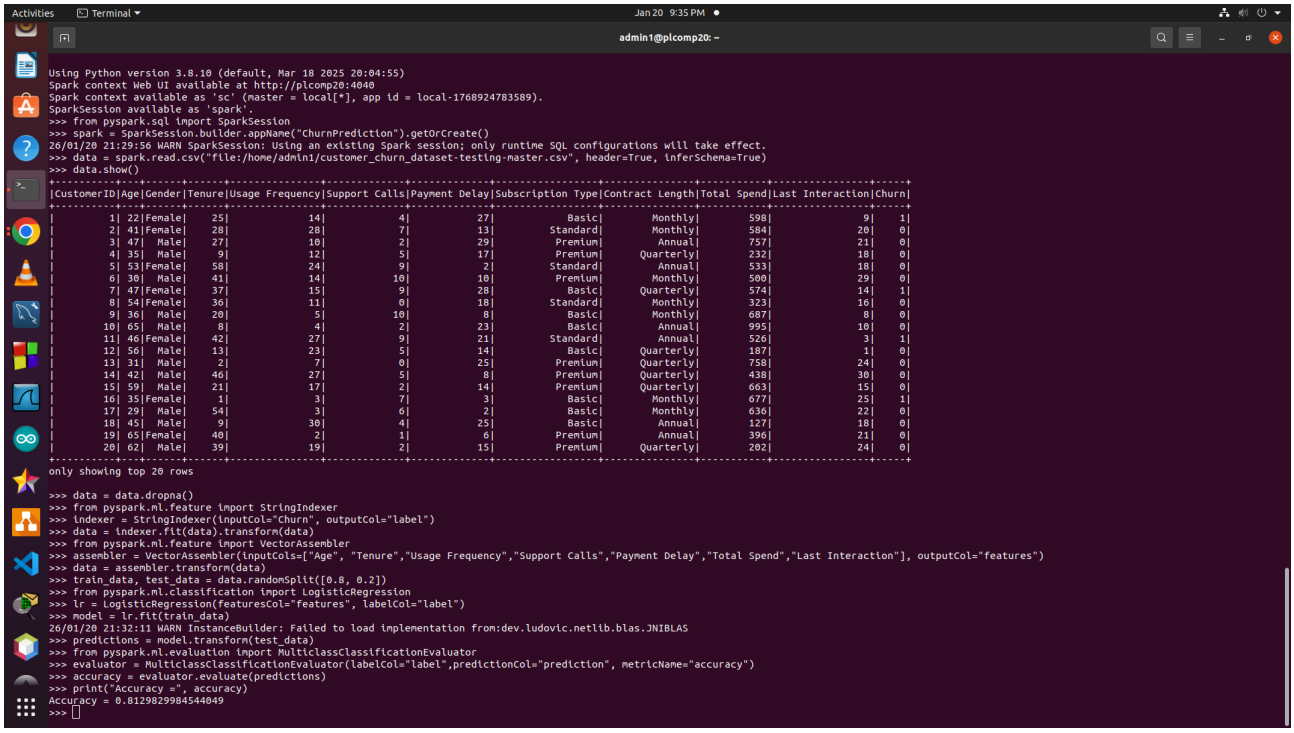

```
Using Python version 3.8.10 (default, Mar 18 2025 20:04:55)
Spark context Web UI available at http://plcomp20:4040
Spark context available as 'sc' (master = local[*], app id = local-1768924783589).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = SparkSession.builder.appName("ChurnPredictction").getOrCreate()
26/01/20 21:29:56 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
>>> data = spark.read.csv("file:/home/admin1/customer_churn_dataset-testing-master.csv", header=True, inferSchema=True)
>>> data.show()
+----------+---+------+------+---------------+-------------+-------------+-----------------+---------------+-----------+----------------+-----+
|CustomerID|Age|Gender|Tenure|Usage Frequency|Support Calls|Payment Delay|Subscription Type|Contract Length|Total Spend|Last Interaction|Churn|
+----------+---+------+------+---------------+-------------+-------------+-----------------+---------------+-----------+----------------+-----+
|         1| 22|Female|    25|             14|            4|           27|            Basic|        Monthly|        598|               9|    1|
|         2| 41|Female|    28|             28|            7|           13|         Standard|        Monthly|        584|              20|    0|
|         3| 47|  Male|    27|             10|            2|           29|          Premium|         Annual|        757|              21|    0|
|         4| 35|  Male|     9|             12|            5|           17|          Premium|      Quarterly|        232|              18|    0|
|         5| 53|Female|    58|             24|            9|            2|         Standard|         Annual|        533|              18|    0|
|         6| 30|  Male|    41|             14|           10|           10|          Premium|        Monthly|        500|              29|    0|
|         7| 47|Female|    37|             15|            9|           28|            Basic|      Quarterly|        574|              14|    1|
|         8| 54|Female|    36|             11|            0|           18|         Standard|        Monthly|        323|              16|    0|
|         9| 36|  Male|    20|              5|           10|            8|            Basic|        Monthly|        687|               8|    0|
|        10| 65|  Male|     8|              4|            2|           23|            Basic|         Annual|        995|              10|    0|
|        11| 46|Female|    42|             27|            9|           21|         Standard|         Annual|        526|               3|    1|
|        12| 56|  Male|    13|             23|            5|           14|            Basic|      Quarterly|        187|               1|    0|
|        13| 31|  Male|     2|              7|            0|           25|          Premium|      Quarterly|        758|              24|    0|
|        14| 42|  Male|    46|             27|            5|            8|          Premium|      Quarterly|        438|              30|    0|
|        15| 59|  Male|    21|             17|            2|           14|          Premium|      Quarterly|        663|              15|    0|
|        16| 35|Female|     1|              3|            7|            3|            Basic|        Monthly|        677|              25|    1|
|        17| 29|  Male|    54|              3|            6|            2|            Basic|        Monthly|        636|              22|    0|
|        18| 45|  Male|     9|             30|            4|           25|            Basic|         Annual|        127|              18|    0|
|        19| 65|Female|    40|              2|            1|            6|          Premium|         Annual|        396|              21|    0|
|        20| 62|  Male|    39|             19|            2|           15|          Premium|      Quarterly|        202|              24|    0|
+----------+---+------+------+---------------+-------------+-------------+-----------------+---------------+-----------+----------------+-----+
only showing top 20 rows

>>> data = data.dropna()
>>> from pyspark.ml.feature import StringIndexer
>>> indexer = StringIndexer(inputCol="Churn", outputCol="label")
>>> data = indexer.fit(data).transform(data)
>>> from pyspark.ml.feature import VectorAssembler
>>> assembler = VectorAssembler(inputCols=["Age", "Tenure","Usage Frequency","Support Calls","Payment Delay","Total Spend","Last Interaction"], outputCol="features")
>>> data = assembler.transform(data)
>>> train_data, test_data = data.randomSplit([0.8, 0.2])
>>> from pyspark.ml.classification import LogisticRegression
>>> lr = LogisticRegression(featuresCol="features", labelCol="label")
>>> model = lr.fit(train_data)
26/01/20 21:32:11 WARN InstanceBuilder: Failed to load implementation from:dev.ludovic.netlib.blas.JNIBLAS
>>> predictions = model.transform(test_data)
>>> from pyspark.ml.evaluation import MulticlassClassificationEvaluator
>>> evaluator = MulticlassClassificationEvaluator(labelCol="label",predictionCol="prediction", metricName="accuracy")
>>> accuracy = evaluator.evaluate(predictions)
>>> print("Accuracy =", accuracy)
Accuracy = 0.8129829984544049
>>>
```